Calculus 12 - Derivative Applications RA (V2)

Name : _____

Teacher's name: _____

Outcome	0	1	2	3	4	5	6	7
Reasoning and Analyzing								

1. For each of the following statements, identify if the statement is **always true**, **sometimes true**, or **never true**. Provide a brief justification.

   (a) If $f'(c) = 0$ and $f''(c) > 0$, then $f(c)$ is a local maximum.

   (b) If $x = c$ is a critical number to $f(x)$, then $f(c)$ is either a maximum or a minimum.

   (c) A function is concave up on (a, b). A linearization centred at $x = c$ where $c \in (a, b)$ gives an under-approximation for values in the interval (a, b).

   (d) If there is an inflection point at $x = c$, then $f''(c) = 0$.

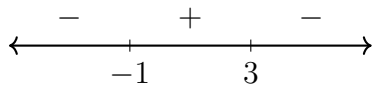
   (e) Intermediate Value Theorem (IVT) is used to find the roots of a function.

2. A student was given the 2nd derivative and asked to find the intervals of concave up and concave down. Which step did he **first** make an error? Explain the error.

$$y'' = (x-3)(x+1)^2$$

$(x-3)(x+1)^2 = 0$ **Step 1**

Possible I.P: $x = 3, -1$ **Step 2**



Step 3

$CU : (-1, 3), CD : (-\infty, -1) \cup (3, \infty)$ **Step 4**

3. In each of the following, one item doesn't belong.

   (a) Which 2nd derivative doesn't belong? Explain.

   i. $y'' = -x^2 - 1$ ii. $y'' = x^2 + 1$ iii. $y'' = \dfrac{1}{x^2+1}$ iv. $y'' = \dfrac{x}{\sqrt{x-1}}$

   (b) Given the function $y = \dfrac{1}{x^2-1}$, which domain interval doesn't belong? Explain.

   i. $(-1, 1)$ ii. $(1, \infty)$

   iii. $(-\infty, -1)$ iv. $(-2, 2)$